# Lecture-04 Score SDE framework

1. Score SDE
2. Score SDE — training and sampling
3. Instantiations of SDEs
4. Rethinking Forward Kernels
5. Fokker-Planck Equation

- In NCSN, it uses increasing noise level $\{\sigma_i\}_{i=1}^L$.
  Data sample $x \sim p_{data}$ is perturbed as
  $$x_i = x + \sigma_i \varepsilon_i, \quad \varepsilon_i \sim N(0, I)$$
- In DDPM, noise incrementally with variance schedule $\{\beta_i\}_{i=1}^L$
  $$x_i = \sqrt{1-\beta_i^2} \cdot x_{i-1} + \beta_i \cdot \varepsilon_i, \quad \varepsilon_i \sim N(0, I)$$

the sequential update from $t \to t+\Delta t$ is

NCSN: $x_{t+\Delta t} = x_t + \sqrt{\sigma_{t+\Delta t}^2 - \sigma_t^2} \cdot \varepsilon_t \approx x_t + \sqrt{\frac{d\sigma_t^2}{dt}\Delta t} \cdot \varepsilon_t$

DDPM: $x_{t+\Delta t} = \sqrt{1-\beta_t}\, x_t + \sqrt{\beta_t} \cdot \varepsilon_t, \quad \varepsilon_t \sim N(0, I)$

$$\approx x_t - \frac{\beta_t}{2}\Delta t \cdot x_t + \sqrt{\beta_t \Delta t}\, \varepsilon_t$$

$\sqrt{1-u} \approx 1 - \frac{u}{2}$ by Taylor expansion

Both two models' noise injection processes follow

$$x_{t+\Delta t} \approx x_t + f(x_t, t) \cdot \Delta t + g(t) \cdot \sqrt{\Delta t} \cdot \varepsilon_t$$

$$\begin{cases} \text{NCSN}: & f(x, t) = 0, \quad g(t) = \sqrt{d\sigma_t^2/dt} \\ \text{DDPM}: & f(x, t) = -\frac{1}{2}\beta_t x, \quad g(t) = \sqrt{\beta(t)} \end{cases}$$

The corresponding transitional:

$$p(x_{t+\Delta t} \mid x_t) := \mathcal{N}(x_{t+\Delta t};\, x_t + f(x_t,t)\Delta t,\, g^2(t)\cdot\Delta t\cdot I)$$

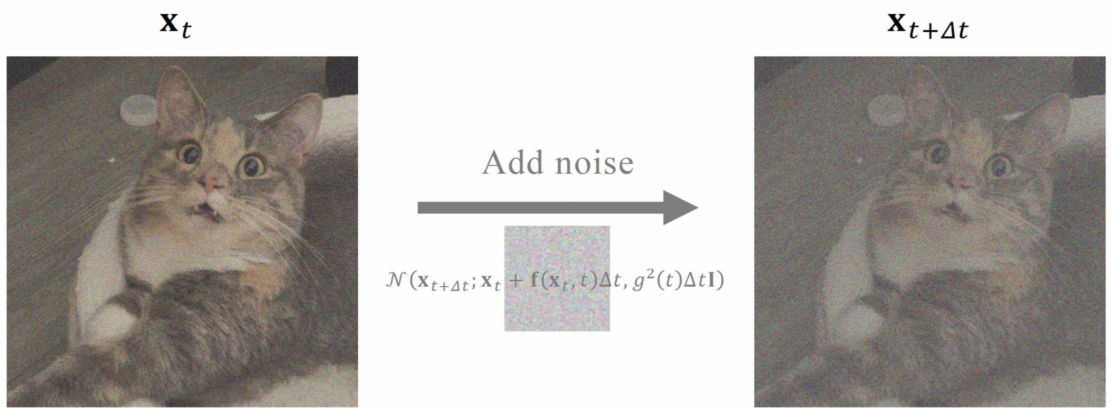



**Figure 4.1: Illustration of the discrete-time noise-adding step.** It adds noise from $t$ to $t + \Delta t$ with mean drift $\mathbf{f}(\mathbf{x}_t, t)$ and diffusion coefficient $g(t)$.

$\Delta t \to 0$: discrete time $\to$ continuous time

SDE evolving forward: $w(t)$: Wiener process / Brownian motion

$$dx(t) = f(x(t),t)\cdot dt + g(t)\, dw(t)$$

as $\Delta t \to 0$, $p(x_{t+\Delta t} \mid x_t) \to$ the corresponding solution of Itô SDE.

$w(t)$:

① any $s<t$. $w(t) - w(s) \sim \mathcal{N}(0, (t-s)\cdot I)$

②. $[t, t+\Delta t]$: $dw(t) = w(t+\Delta t) - w(t) \Leftrightarrow dw(t) \sim \mathcal{N}(0, dt\, I)$

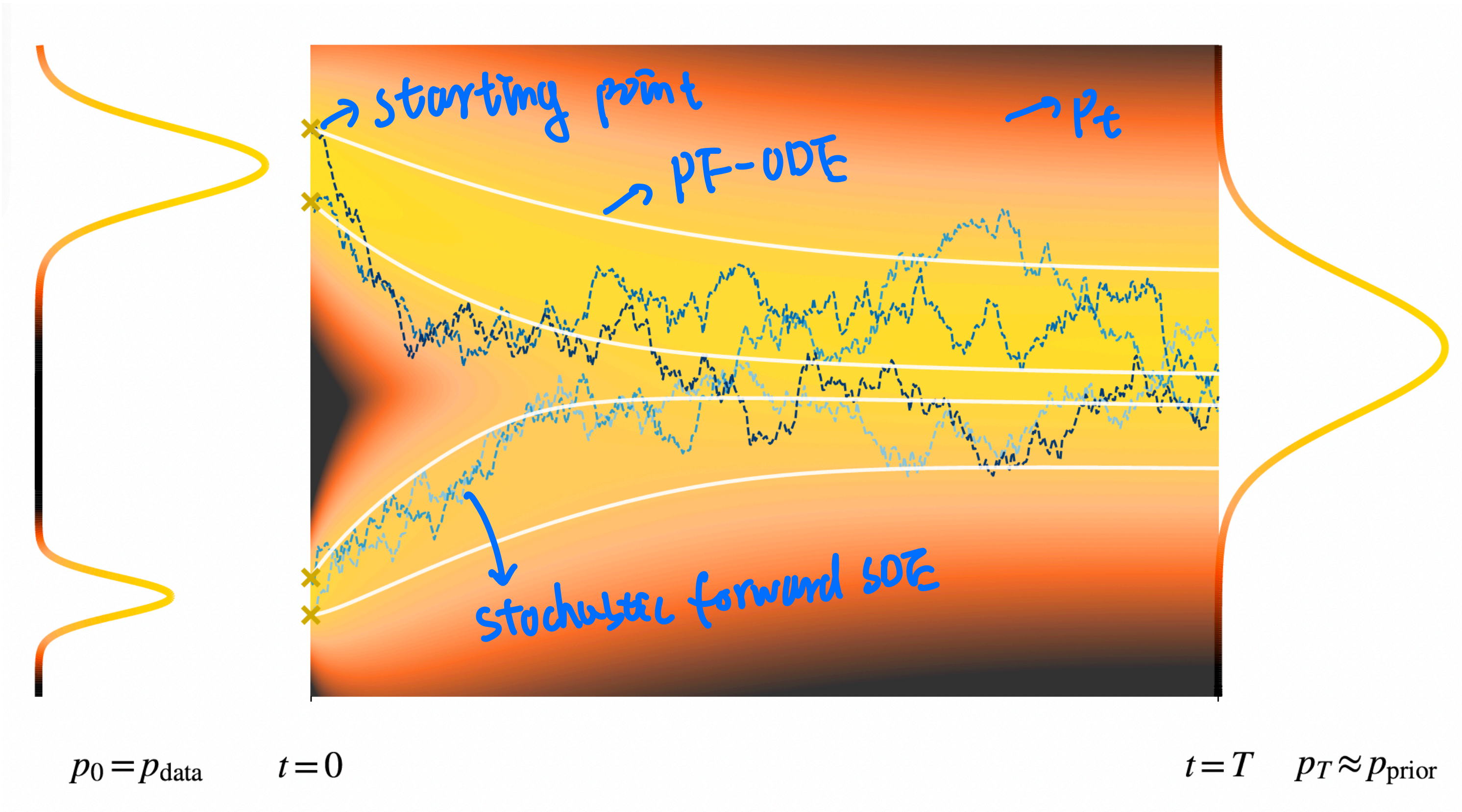
③. initial state: $w(0) = 0$ almost surely

④. $w(t) - w(s)$ is independent of the past

⑤. $t \mapsto w(t)$ almost surely continuous but nowhere differentiable.

$$\begin{cases} x(t+\Delta t) - x(t) \approx dx(t) \\ \Delta t \approx dt \\ \sqrt{\Delta t}\,\varepsilon_t \sim \mathcal{N}(0, \Delta t\, I) \simeq dw(t) \end{cases}$$

# From data to noise: forward-time SDEs




## Forward SDE:

$$\boxed{dx(t) = f(x(t),t)\cdot dt + g(t)\, dw(t),\ x(0) \sim p_{data}}$$

It induced perturbation kernel:

$p_t(x_t \mid x_0)$, $x_0 \sim p_{data}$, drift: $f(x,t)$

Assume there $f(x,t) = f(t)\cdot x$ : affine, then:

$$p_t(x_t \mid x_0) = \mathcal{N}(x_t; m(t), p(t)\cdot I_D) \text{ with}$$

$$m(t) = \exp\left(\int_0^t f(u)\,du\right)\cdot x_0$$

$$p(t) = \int_0^t \exp\left(2\int_s^t f(u)\,du\right)\cdot g^2(s)\,ds$$

with

initial condition: $\begin{cases} m(0) = x_0 \\ p(0) = 0 \end{cases}$

NCSN and DDPM fall into this affine-drift setting.

For deterministic time varying function $y_t$ (an ODE)

$$\frac{dh(y_t,t)}{dt} = \frac{\partial h}{\partial t} + \nabla_y h \cdot \frac{dy_t}{dt}$$

$$\frac{dy}{dt} = f(y,t)$$

$$h(y_t,t) : \mathbb{R}^D \times \mathbb{R}^1 \to \mathbb{R}^D$$

$\nabla_y h$ is the Jacobian matrix of $h$

$$\nabla_y h = \begin{bmatrix} \frac{\partial h_1}{\partial y_1} & \frac{\partial h_1}{\partial y_2} & \cdots & \frac{\partial h_1}{\partial y_D} \\ \frac{\partial h_2}{\partial y_1} & \frac{\partial h_2}{\partial y_2} & \cdots & \frac{\partial h_2}{\partial y_D} \\ \vdots & \vdots & \ddots & \\ \frac{\partial h_D}{\partial y_1} & \frac{\partial h_D}{\partial y_2} & \cdots & \frac{\partial h_D}{\partial y_D} \end{bmatrix}$$

$$y_t = \begin{bmatrix} y_1(t) \\ y_2(t) \\ \vdots \\ y_D(t) \end{bmatrix} \qquad h = \begin{bmatrix} h_1(t) \\ h_2(t) \\ \vdots \\ h_D(t) \end{bmatrix}$$

But what if $x_t$ is a stochastic process driven by SDE:

$$dx_t = f(x_t,t)\,dt + g(t)\cdot dw(t)$$

What SDE does the process $h(x_t,t)$ satisfy?

$$\frac{dh}{dt} = \frac{\partial h}{\partial t} + \nabla_x h \cdot \boxed{\frac{dx_t}{dt}} \Leftrightarrow dh = \frac{\partial h}{\partial t}\cdot dt + \nabla_x h \cdot dx_t$$

problematic

$dw_t = O(\sqrt{dt}) \Rightarrow (dw_t)^2 = dt$ : second-order terms in $dw_t$ do not vanish.

$$\begin{cases} (dt)^2 \approx 0 \\ (dw_t)^2 \approx dt \times 0. \end{cases}$$

Example: Consider $x_t : \mathbb{R} \to \mathbb{R}$

$h(x_t) = x_t^2$ with $dx_t = \sigma\, dw_t$, $\sigma > 0$ use classic chain rule: $dh = 2x_t\, dx_t = 2x_t \sigma\, dw_t$

But, $\mathbb{E}[x_t^2] = \sigma^2 \cdot t$.

If it is correct, then $\mathbb{E}\,dh = 2\sigma\, \mathbb{E}[x_t dw_t] = 2\sigma \cdot \mathbb{E}[x_t] \cdot \underbrace{\mathbb{E}[dw_t]}_{=0} = 0.$

$$x_t = e^{\int_0^t f(s)ds}\left[x_0 + \int_0^t \frac{e^{-\int_0^s f(r)dr} g(s)}{\phi(s)} dW(s)\right.$$

$$Z_t := \int_0^t \phi(s)\, dW_s$$

$$A(t) = e^{\int_0^t f(s)ds}$$

$$X_t = m(t) + A(t)\cdot Z_t$$

$$\begin{aligned} P(t) &= \mathbb{E}[(X_t - m(t))\cdot(X_t - m(t))^T \mid x_0] \\ &= \mathbb{E}[A(t)^2 \cdot Z_t Z_t^T \mid x_0] \\ &= e^{2\int_0^t f(s)ds} \cdot \mathbb{E}[Z_t Z_t^T] \end{aligned}$$

Define $\lambda(t) = \exp\left(\int_0^t f(u)\,du\right)$

Let $y(t) = \lambda(t)^{-1} x(t)$